QPV-BB84 under stress: 40 scenarios

Nov 4, 2026 · @QubitMan

The short version.

- An honest prover always passes.
- Cheaters who don't share entanglement pass at most about 85% of rounds, so a few dozen rounds catch them.
- Cheaters who share just *one* entangled pair of particles pass every round. That is the protocol's biggest weakness.
- If more than half the photons get lost on the way, the protocol stops working.

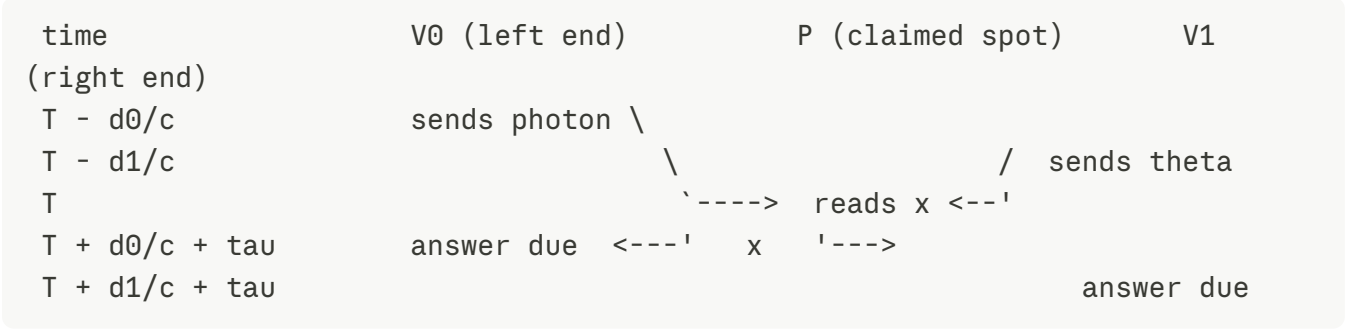
The rest of this document goes through each situation one at a time.

How the protocol works

Picture a straight road. Verifier **V0** stands at one end and verifier **V1** at the other. Somewhere in between, a **prover** claims to be standing at a spot we call **P**.

Each round goes like this:

1. **Before the round,** the two verifiers secretly agree on two random bits: a secret bit **x**, and a "basis" bit **θ** that says how x will be hidden.
2. **V0 sends a single photon (a qubit)** with x hidden inside it, using basis θ. Without knowing θ, nobody can read x reliably. **V1 sends θ** as an ordinary message.
3. **The timing is set so both arrive at P at exactly the same moment,** called T. The verifier farther away simply sends earlier.
4. **The prover reads the photon using θ,** gets x, and sends x back to both verifiers.
5. **Each verifier checks two things:** is the bit right, and did it come back as fast as light allows? They give the prover a small allowance **τ** (the tolerance) for its own processing time.



Here d0 and d1 are the distances from P to each verifier, and c is the speed of light.

Why this proves location. To answer correctly you need the photon and θ together, in the same place, at the same time. That only happens at P. Anyone standing elsewhere has to answer at least one verifier before they could possibly have both pieces, so they are forced to guess.

How precise it is. Because of the allowance τ, the protocol can only pin the prover down to within c × τ / 2 of P. That is about **15 cm for every nanosecond of allowance**.

Words used in this document

Term	Meaning
Qubit	The single photon carrying the hidden bit
Basis	The "setting" used to hide or read the bit. Read with the wrong setting, you get a random answer
Breidbart basis	A clever in-between setting that reads the bit correctly about 85% of the time without knowing θ
EPR pair	Two entangled particles, one held by each of two people. Lets them "teleport" a qubit
Collusion	Two or more attackers working together from different places
Tolerance τ	The extra time allowed for the prover to process the photon

What each scenario covers: what happens and when; how often the verifiers accept; whether security still holds and which assumption is being tested; the best known attack and where it comes from; how to defend against it and what that costs; and how to try it in the QPV Workbench simulator.

About the numbers. A number with a paper next to it comes from that paper. *Worked out here* means a short calculation you can check yourself from the rules above. *Open* means researchers have no settled answer yet.

Trying it yourself. Every scenario below is in the QPV Workbench under **Scenarios**. Click one to run it, then press *Play 100 rounds* to see the pass rate. Or click *Edit* to open it in the scenario builder. There, a scenario is a standard setup plus a list of "twists" (move someone, slow a link, put someone in the middle, leak a secret, and so on), so you can change any detail or build new cases the same way.

How to run every scenario

Every case in this document is a ready-made scenario in the QPV Workbench. There are three ways to run one.

1. **In your browser, nothing to install:** open the link in the table below. The scenario loads and runs one round. Press *Play 100 rounds* in the right-hand panel to see the pass rate, and how the other rounds ended.
2. **On your own computer:** download the code and run `./run.sh` (it installs everything, builds the interface once and opens your browser). Add a scenario name to go straight to it, for example `./run.sh d1_teleport`.
3. **From the command line:** `qpv scenario <name> --rounds 500` runs a scenario 500 times and prints the pass rate and how each verifier's rounds ended. `qpv scenarios` lists all the names.

To change a scenario, click **Change it…** on its card, or **Edit** in the Scenarios list. That opens the scenario builder, where each case is a standard setup plus a short list of "twists" you can edit, add or remove. *Save* keeps your version under **My scenarios**.

Case	Scenario	Who answers	Open in the browser	Command
A1	Prover exactly in the middle	Honest prover	open	`qpv scenario a1_midway`
A2	Prover closer to one verifier	Honest prover	open	`qpv scenario a2_off_centre`
A3	Claim outside the verifiers	Cheater	open	`qpv scenario a3_outside`
A3	Claim off to the side	Cheater	open	`qpv scenario a3_off_line`
A4	Prover right at a verifier	Honest prover	open	`qpv scenario a4_at_verifier`
A5	3D: four verifiers around the prover	Honest prover	open	`qpv scenario a5_3d`
A7	Prover 4 cm off, inside the precision	Honest prover	open	`qpv scenario a7_within`
A7	Prover 6 cm off, beyond the precision	Honest prover	open	`qpv scenario a7_beyond`

Case	Scenario	Who answers	Open in the browser	Command
B1	Prover moved 3 cm (10% of light speed for 1 ns)	Honest prover	open	`qpv scenario b1_moving`
B2	Prover takes 0.8 ns to respond	Honest prover	open	`qpv scenario b2_slow_prover`
B3	V1's clock is 1 ns ahead	Honest prover	open	`qpv scenario b3_clock_offset`
B4	V1 uses fibre and adjusts for it	Honest prover	open	`qpv scenario b4_fibre_honest`
B4	Fibre adjustment exploited by a cheater	Cheater	open	`qpv scenario b4_fibre_cheater`
B5	Noisy channel (15% errors)	Honest prover	open	`qpv scenario b5_noise`
B6	60% of photons lost	Honest prover	open	`qpv scenario b6_loss`
B7	The laser sends two photons	Cheater	open	`qpv scenario b7_two_photons`
B8	Why many rounds matter	Cheater	open	`qpv scenario b8_many_rounds`
C1	Lone cheater on V1's side	Cheater	open	`qpv scenario c1a_liar_theta_side`
C1	Lone cheater on V0's side	Cheater	open	`qpv scenario c1b_liar_qubit_side`
C2	Two cheaters guessing the basis	Cheater	open	`qpv scenario c2_guess_basis`
C2	Two cheaters, best possible strategy	Cheater	open	`qpv scenario c2_breidbart`
C3	Eavesdropper reads and resends the photon	Honest prover	open	`qpv scenario c3_intercept`
C4	Cheaters who just guess	Cheater	open	`qpv scenario c4_guess`
D1	Teleportation with one entangled pair	Cheater	open	`qpv scenario d1_teleport`

Case	Scenario	Who answers	Open in the browser	Command
D3	Teleportation chain in 2D (XOR basis)	Cheater	open	`qpv scenario d3_chain_2d`
D4	Poor entanglement (50% noise)	Cheater	open	`qpv scenario d4_noisy_low`
D4	Good entanglement (10% noise)	Cheater	open	`qpv scenario d4_noisy_high`
E1	Cheater standing at the claimed spot	Cheater	open	`qpv scenario e1_impostor`
E3	Someone in the middle flips the basis	Honest prover	open	`qpv scenario e3_tamper`
E3	Someone in the middle holds messages for 2 ns	Honest prover	open	`qpv scenario e3_delay`
E4	Cheater right next to V0	Cheater	open	`qpv scenario e4_near_verifier`
E5	Honest prover and cheater at the same spot	Cheater	open	`qpv scenario e5_two_at_claim`
F1	Cheaters say 'no photon' when they guess wrong	Cheater	open	`qpv scenario f1_selective_loss`
F3	Cheater with a straight route past a 12 m fibre detour	Cheater	open	`qpv scenario f3_fibre_detour`
F4	The basis leaks to a cheater	Cheater	open	`qpv scenario f4_leaked_basis`

Not every case has a scenario of its own. A6 is C1; C5, C6 and C7 are shown by C2 with *Play 100 rounds*; D2, D5, E2, E6 and F2 are discussions rather than simulations.

New scenarios

These twelve were added to the gallery later. They cover cases the table above could only describe: tests over many rounds, answers with a key, weak lasers, a walking prover and a protocol that copes with lost photons.

Case	Scenario	Who answers	Open in the browser	Command
B1	Walking prover, verifiers don't follow it	Honest prover	open	`qpv scenario b1_walking_untracked`
B1	Walking prover, verifiers follow its path	Honest prover	open	`qpv scenario b1_walking_tracked`
B5	2% errors, judged over 360 rounds	Honest prover	open	`qpv scenario b5_threshold_honest`
B5	The same 360-round test against the best cheaters	Cheater	open	`qpv scenario b5_threshold_cheaters`
B7	Weak laser (0.5 photons per pulse), honest prover	Honest prover	open	`qpv scenario b7_weak_laser_honest`
B7	Weak laser, cheaters wait for two-photon pulses	Cheater	open	`qpv scenario b7_weak_laser_split`
B7	Detector false clicks (10%)	Honest prover	open	`qpv scenario b7_dark_counts`
B8	88 rounds against the best cheaters	Cheater	open	`qpv scenario b8_88_rounds`
E1	Impostor at the claimed spot, answers need a key	Cheater	open	`qpv scenario e1_keyed_impostor`
E1	Genuine prover with the key	Honest prover	open	`qpv scenario e1_keyed_honest`
F2	Four bases: honest prover with 50% loss	Honest prover	open	`qpv scenario f2_four_bases_honest`
F2	Four bases: selective 'no photon' cheaters caught	Cheater	open	`qpv scenario f2_four_bases_cheaters`

Showing them to others

- **Present them one by one:** open this link, or click **Scenarios**, pick all or one section and press **Present**. Each case fills the screen, plays one round, then 100, and shows the measured pass rate next to the expected one. The arrow keys move between cases; Esc leaves.
- **Check them all at once: Report…** in the Scenarios list runs every case and saves a table of measured against expected pass rates, as Markdown, CSV or a page you can print to PDF. From the command line: `qpv report --out report.html`.
- **See how a setting changes the result: Chart…** on a scenario's card plots its pass rate as one setting changes (noise, loss, photons per pulse, false clicks, distance and more), with a dashed theory line where a simple formula exists.
- **Share your own version:** after changing a scenario in the builder, **Copy link** gives a link that opens exactly that version.
- **How many rounds are needed:** `qpv rounds --honest 0.98 --cheat 0.854` gives the fewest rounds, and the pass mark, that separate an honest prover from a cheater (here 360 rounds with 337 right, for a one-in-a-million chance of either mistake).

A. Honest prover, perfect conditions

With perfect equipment, an honest prover passes every time. The real question is *which spot* the protocol actually proves. On a line, it proves "within $c \times \tau / 2$ of P", but only if P is between the two verifiers.

#	Situation	Honest prover passes?	Can a cheater fake it?	What it tests
A1	P exactly in the middle	Always	Not alone (colluders: see C and D)	Speed of light
A2	P closer to one verifier	Always	Same as A1	Speed of light
A3	P outside the stretch between the verifiers, or off to the side	Always	**Yes, every time**, from the nearest point between them	Geometry
A4	P right at a verifier	Always	Only from inside that verifier's building	Trusted verifier sites
A5	Flat area (2D) or space (3D)	Always	Not if P is surrounded by enough verifiers	Geometry

#	Situation	Honest prover passes?	Can a cheater fake it?	What it tests
A6	Prover claims the wrong spot	Only if the error is under c × τ / 2	A liar passes 50% or 85% of rounds, depending on side	Speed of light, no-cloning
A7	Timing errors	Yes, if τ is big enough	The bigger τ, the blurrier the location	Equipment precision

A1. P in the middle

Both verifiers are the same distance away, so they send at the same moment. A lone cheater standing anywhere else either misses a deadline or has to answer one verifier blind (see A6).

A2. P closer to one verifier

Each verifier sends early by its own travel time: the farther one sends first. Its deadline is later by the same amount. Security doesn't change: on a line, the protected zone is still c × τ / 2 either side of P.

```
       V0 ---------------- 3 m ---- P ---- 1 m ---- V1
sends  T - 10 ns                                  T - 3.3 ns
meet                              T
answer due at V0: T + 10 ns + tau        at V1: T + 3.3 ns + tau
```

In a flat area or in space, placement does matter. If the verifiers are almost in a line with P, small timing errors stretch the zone sideways. GPS has the same problem.

A3. P outside the stretch, or off to the side

Outside. Say P is beyond V1. A cheater standing right next to V1 gets V0's photon and V1's θ at the same moment, earlier than P would. It reads x correctly and answers both verifiers early. It passes every round. *(Worked out here.)*

Off to the side. Two verifiers on a line can't tell "P" from "the point on the line level with P". That point is closer to both verifiers, so a cheater there passes every round. Two verifiers only check the position *along* their line.

- **Fix:** keep claims inside the area surrounded by the verifiers. On a flat area that means a third verifier off the line. Cost: one more verifier, synchronised with the others.

A4. P right at a verifier

If P is at V0, then V0 sends the photon and expects an answer almost instantly. The protocol then only proves "someone is right next to V0", so it is only as safe as V0's own building. It is allowed, but rarely useful.

A5. 2D and 3D: how many verifiers you need

- **On a flat area:** at least 3 verifiers, not in a line.
- **In space:** at least 4, not all on one flat plane.
- **P must be strictly inside the shape they form** (the "convex hull"). If P is outside it, the closest point of that shape is nearer to *every* verifier, so a cheater there passes, as in A3. If P is inside, no other point is that close to all of them at once.
- **Where this comes from:** this rule was worked out for classical (non-quantum) position checking by Chandran, Goyal, Moriarty and Ostrovsky (CRYPTO 2009).
- **Caveat:** most quantum security proofs only cover the simple line with two verifiers. Treat 2D and 3D security as a stated assumption.

A6. Claiming the wrong spot

- **Honest mistake:** if the error is less than $c \times \tau / 2$, the prover still passes. Beyond that, an honest prover who waits for both signals answers too late and always fails.
- **Deliberate lie, standing on V1's side of P:** the liar must answer V0 before anything from V0 could reach it. That answer is a coin flip, so it passes **50%** of rounds, and only 0.5^n over n rounds. *(Worked out here.)*
- **Deliberate lie, standing on V0's side of P:** the liar gets the photon early but must answer V1 before θ arrives. Reading the photon in the Breidbart basis gives the right bit about **85%** of the time. That is the best possible, because one liar is just a special case of the two cheaters in C5.
- **Note on the Workbench:** its "displaced prover" waits for everything and answers late, so it fails every round. A smarter liar would do better, as above.

A7. How precise can it be?

The allowance τ must cover the prover's processing time plus timing jitter, or honest provers start failing. The location is then only known to within $c \times \tau / 2$:

Allowance τ	Location known to within
10 ns	±1.5 m
1 ns	±15 cm

Allowance τ	Location known to within
100 ps	±1.5 cm
10 ps	±1.5 mm

Trying section A in the Workbench

- **In the Scenarios gallery:** A1, A2, both A3 cases, A4, A5 and both A7 cases. Click one to run it, or *Edit* to change it.
- **Build your own:** the twists "Move a party" and "Move the claimed position" cover A2 to A4 and A7. "Change how a party answers" turns the prover into a cheater (A3).
- **See the safe zone:** in 2D, the Map view's light-cone check paints where a lone cheater could pass.
- **Smart liars (A6):** see C1 in the gallery.

B. Honest prover, real-world conditions

Real equipment is never perfect. Every imperfection does one of two things: it adds extra waiting time, which makes the location blurrier, or it adds errors, which make the honest prover look more like a cheater. The two hardest problems are lost photons (B6) and lasers that sometimes send two photons instead of one (B7).

#	Problem	Honest prover still passes?	Effect on security	What it tests
B1	Prover is moving	Yes, at everyday speeds	None within a round; each round must aim at the predicted spot	Prover standing still
B2	Prover is slow to respond	Yes, if the allowance τ covers it	Location gets blurrier	Instant processing
B3	Verifier clocks disagree	Yes, while the error stays inside τ	The checked spot shifts	Synchronised clocks
B4	Signals travel slower than light (fibre)	Yes, if send times are adjusted	Cheaters using faster routes gain time	Signals at light speed

#	Problem	Honest prover still passes?	Effect on security	What it tests
B5	Noisy channel	Mostly; it makes some mistakes	Above 14.6% errors you can't tell honest from cheating	Perfect channel
B6	Lost photons	Only on rounds that arrive	Breaks completely if half or more are lost	Perfect channel
B7	Two photons sent instead of one	Yes	Those rounds are free wins for cheaters	Single-photon source
B8	Not enough rounds	—	About 88 rounds needed on a perfect line, about 1,750 with 2% noise	Many rounds

B1. A moving prover

What matters is how far the prover moves during its own processing time, which lasts τ:

Speed	Moves during 1 ns	Frequency shift (Doppler)	A problem?
Walking, 1.5 m/s	1.5 nanometres	0.0000005%	No
Car, 30 m/s	30 nanometres	0.00001%	No
Plane, 250 m/s	250 nanometres	0.00008%	No
10% of light speed	3 cm	10%	Yes

- **Between rounds:** the claim becomes a path, and each round aims at where the prover should be at that moment. The prediction must be accurate to within $c \times \tau / 2$, or honest rounds fail like a wrong claim (A6).
- **At extreme speeds:** a 10% frequency shift moves the light outside the detector's range, and the photon's "settings" look rotated from the prover's point of view. The protocol would need to correct for this. *Open:* I don't know of a standard analysis for very fast provers.
- **Workbench:** positions are fixed during a run. The gallery's "B1 · Prover moved 3 cm" shows where a very fast prover would be when it answers; for a whole path, run one scenario per position.

B2. A prover that is slow to respond

Detectors and electronics take time, say δp. The verifiers must make the allowance τ at least that big, which blurs the location to c × τ / 2. For example, 10 ns of electronics means ±1.5 m. Cheaters get the same allowance.

- **Fix:** fast detectors and dedicated hardware logic, a few ns. Cost: specialised equipment.
- **Workbench:** the prover's `processing_ns` against `tolerance_ns`, or sweep the `processing` setting.

B3. Verifier clocks that disagree

If V1's clock runs δ ahead, V1 sends δ too early and also stops waiting δ too early. The meeting point slides c × δ / 2 towards V0, and the honest prover has δ less time to reach V1. Once δ plus the processing time is more than τ, honest provers fail.

- **Fix:** keep the clocks in step far more tightly than τ, for example with White Rabbit, a timing system designed for better than a nanosecond. Cost: dedicated timing links.
- **Workbench:** set a verifier's `clock.offset_ns` or `clock.drift_ppm`, or sweep `drift`. With τ = 0.5 ns, an offset of 0.2 ns still passes and 1 ns fails.

B4. Signals slower than light

Light in optical fibre travels at about 2/3 of its speed in air. Verifiers can allow for this by sending earlier and waiting longer, so the honest prover passes. But a cheater can relay the signal through open air at full speed and save time. For every ℓ metres of fibre, the cheater gains about ℓ/c of spare time, which blurs the location by about ℓ/2. So 3 m of fibre turns ±7.5 cm into roughly ±1.5 m. *(Worked out here.)*

- **Fix:** send the challenges through open air in a straight line, or keep fibre very short and count its delay inside τ. Cost: optics, alignment and sensitivity to weather.
- **Workbench:** set a link's `speed_mps` or `latency_ns`. Without adjusting, the honest prover fails. In the gallery, "B4 · V1 uses fibre and adjusts for it" and "B4 · Fibre adjustment exploited by a cheater" show both sides; the builder's "Change a verifier's links" twist sets the speed and whether the verifier adjusts its timing.

B5. A noisy channel

Noise flips some of the honest prover's answers, say a fraction ε. Cheaters without entanglement get at least 14.6% wrong (C5). So the verifiers accept if the error rate is below a cut-off somewhere between ε and 14.6%. If the equipment's own error rate approaches 14.6%, honest provers and cheaters look the same. Against cheaters *with* entanglement (D1), no cut-off helps at all.

- **Caveat:** Tomamichel et al. (2013) give the exact maths for choosing the cut-off. 14.6% alone is only the rule of thumb.
- **Workbench:** set `noise.depolarizing` on the photon link, or sweep `noise`. A depolarising strength of q gives an error rate of q/2.

B6. Lost photons

Some photons never arrive: a fraction 1 − η is lost, where η is the fraction that gets through. The verifiers have two options, and both have a hard limit. *(Both worked out here.)*

- **Allow the prover to say "nothing arrived":** cheaters then guess the basis and claim "nothing arrived" whenever they guessed wrong. That looks like 50% loss with no mistakes (F1). So if half or more of the photons are lost anyway, cheaters are invisible.
- **Require an answer every round:** the honest prover has to guess when nothing arrives, and gets (1 + η)/2 right. To beat cheaters' 85%, at least 71% of photons must arrive. Only very short, clean links manage that.
- **Fix:** loss-tolerant protocols (F2). Cost: a different protocol, often with stronger assumptions about the cheaters.
- **Workbench:** set `loss_prob` or `attenuation_db_per_km` on the photon link. The prover then simply never answers (gallery: "B6 · 60% of photons lost"). The twist "Let the prover report 'no photon'" allows loss reports.

B7. Two photons instead of one, and detector glitches

- **Two photons:** cheap lasers sometimes send two or more photons instead of one, with probability 1 − e^(−μ)(1 + μ), roughly μ²/2 for small μ, where μ is the laser's average number of photons per pulse. With two copies, each cheater keeps one, waits for θ, and reads it correctly: a guaranteed win. Combined with loss, cheaters can block single-photon rounds and only answer two-photon ones, as in attacks on quantum key distribution. Lim et al. (2016) study position checking with this kind of laser.
- **Dark counts:** a detector sometimes clicks when no photon arrived, giving a random bit. This adds a little to the error rate.
- **Double clicks:** when both detectors click, record a random bit rather than throwing the round away. Otherwise a cheater could choose which rounds count.
- **Fix:** true single-photon sources, or decoy-pulse tricks borrowed from quantum key distribution. Cost: more complex sources and fewer rounds per second.
- **Workbench:** open "B7 · The laser sends two photons": the "Photons per pulse" twist plus the photon-splitting cheaters. Dark counts and double clicks are not modelled.

B8. How many rounds are enough?

- **Perfect line, cheaters without entanglement:** they pass all n rounds with probability at most 0.854^n (proven by Tomamichel et al. 2013). For a one-in-a-million chance you need **88 rounds**.
- **With 2% noise:** honest provers get 98% right and cheaters at most 85.4%. Setting the cut-off halfway, about **1,750 rounds** keep both kinds of mistake below one in a million. *(Worked out here with a standard statistics rule, Hoeffding's inequality; it is an estimate, not a proof.)*
- **Cheaters with entanglement:** no number of rounds helps (D1).
- **Workbench:** run an Experiment with `runs` set to n. Results come with 95% confidence intervals.

C. Cheaters without entanglement

Cheaters who don't share entanglement can never pass more than about **85%** of rounds, exactly $\cos^2(\pi/8) \approx 0.854$. That is proven, even if they have perfect quantum computers, quantum memory and can talk to each other as fast as light allows (Tomamichel, Fehr, Kaniewski and Wehner, 2013). Over many rounds, 85% shrinks fast: 0.854^n.

#	Who cheats, and how	Passes per round	Chance of passing 50 rounds	Source
C1a	One cheater on V1's side of P	50%	about 1 in a quadrillion	Worked out here
C1b	One cheater on V0's side of P	85%	about 4 in 10,000	Worked out here; best possible (Tomamichel et al.)
C2	Two cheaters, one on each side, guessing the basis	75%	about 6 in 10 million	Well known
C2*	Two cheaters, using the Breidbart basis	85% (the best possible)	about 4 in 10,000	Tomamichel et al. 2013
C3	Grab the photon, read it, pass a copy on to the real prover	Real prover still passes 75%	—	Well known from quantum key distribution
C4	Pure guess of the bit	50%	about 1 in a quadrillion	—

#	Who cheats, and how	Passes per round	Chance of passing 50 rounds	Source
C7	Cheaters with quantum memory, but no entanglement	At most 85%	at most 4 in 10,000	Tomamichel et al. 2013

C1. One cheater, standing in the wrong place

How well a lone cheater does depends on which side of P it stands, because the two verifiers send different things. *(Worked out here.)*

- **On V1's side:** it gets θ early, but its answer to V0 must leave before anything from V0 could reach it. So its answer to V0 can't depend on x. It is right half the time.
- **On V0's side:** it gets the photon early, but must answer V1 before θ arrives. Reading the photon in the Breidbart basis, a setting halfway between the two possible bases, gives the right bit about 85% of the time whatever θ is. It sends that bit to both verifiers. This is the best possible: one cheater is just a weaker version of two.

```
 V0 ----- E ---------- P ---------- V1      (cheater E on V0's side)
 photon reaches E early -> E reads it in the Breidbart basis -> answer to V1
must leave now
 theta reaches E only after that answer had to leave
```

C2. Two cheaters working together

One cheater sits between V0 and P, the other between P and V1. Between them they have everything, but not in one place.

1. The first cheater gets the photon and reads it straight away, either guessing the basis or using the Breidbart basis.
2. The second cheater gets θ.
3. They swap what they know at light speed: the first sends its result, the second sends θ.
4. Each answers its own verifier on time.

Guessing the basis: right half the time, a coin flip otherwise, so 75% overall. **Breidbart basis:** about 85%, the best that is possible.

C3. Grab, read and pass it on

A cheater on the road between V0 and P reads the photon in a guessed basis, then sends a fresh photon with what it found on to the real prover. When it guessed wrong (half the time), the real prover's answer becomes random. So the real prover gets 25% wrong, just like an eavesdropper in quantum key distribution. This is sabotage, not impersonation: it doesn't let the cheater pass as the prover.

C4. Guessing the basis versus guessing the bit

Strategy	Gets the bit right
Just guess the bit, ignore the photon	50%
Guess the basis, read the photon with it	75%
Read the photon in the Breidbart basis	about 85%
Wait for θ, then read it (only possible at P)	100%

C5. The proven limit

Tomamichel and colleagues treat the protocol as a game against cheaters who may do *anything* to the photon: split it, store it, run quantum computations, and send each other quantum or ordinary messages. Without entanglement shared beforehand, they win a round at most $\cos^2(\pi/8) \approx 85.4\%$ of the time. They also prove that n rounds are won at most 0.854^n of the time, which is the number used in B8.

C6. Over many rounds

Rounds	Chance cheaters pass every one
10	21%
20	4.2%
50	0.037%
88	about 1 in a million
100	about 1 in 7.5 million

C7. Quantum memory, but no entanglement

Memory on its own doesn't help within a round; the 85% limit already covers it. The danger is *between* rounds. Cheaters with memory and a quantum link can share entangled pairs in advance, and then become the much stronger cheaters of section D. So "no shared entanglement" is really an assumption about cheaters' equipment: no long-lasting quantum memory, or no quantum link between them.

How to defend against section C

Run many rounds: 88 rounds push cheaters without entanglement below one in a million (B8). Cost: a longer check, and the honest error rate must stay well below 14.6%.

Trying section C in the Workbench

- **In the Scenarios gallery:** C1 (both sides), C2 (guessing and Breidbart), C3 eavesdropper and C4 guessing. B8 shows what many rounds do.
- **Build your own:** set "Who answers" to the unentangled attack and pick a measurement policy, or use the twist "Change how a party answers" with *measure on arrival* or *guess first*. "Put someone in the middle of a path" with *read and resend* gives C3.
- **Compare:** the Compare tab plays the honest prover against every attack, with the theory next to it.

D. Cheaters with entanglement

One shared entangled pair is enough to beat the basic protocol every single time. With unlimited entanglement, *every* quantum position check can be beaten (Buhrman et al., 2014). So research doesn't aim for perfect security. It aims to make cheating need so much entanglement that it isn't practical.

D1. The teleportation trick, step by step

This attack was found by Kent, Munro and Spiller (2011) and by Lau and Lo (2011). Two cheaters stand either side of P, and before the round they share one entangled pair of particles: one half each.

1. **The first cheater teleports the photon right away.** As soon as the photon arrives, it does a joint measurement (a "Bell measurement") on the photon and its half of the pair. This gives it two ordinary bits, s and t. The photon's state now sits in the *other* cheater's particle, slightly scrambled in a way that s and t describe.
2. **The second cheater reads it with the right setting.** θ reaches the second cheater early, so it reads its particle in basis θ straight away and gets a bit y.

3. **The scrambling only flips the bit.** Depending on θ, y equals x, or x flipped by s or by t. Which bit to undo is just ordinary arithmetic.
4. **They swap notes at light speed.** The first sends s and t, and the second sends θ and y.
5. **Both work out x and answer.** Each answer arrives exactly when an honest prover's would.

Cost: one entangled pair per round and a few ordinary bits. Success: **100%**. The trick works because teleporting moves the photon *before* anyone knows θ, and the clean-up afterwards is ordinary information that can travel separately.

D2. With unlimited entanglement, nothing is safe

- **Without quantum physics:** any ordinary (classical) position check can be beaten by cheaters working together (Chandran, Goyal, Moriarty and Ostrovsky, 2009). That is why quantum versions were invented.
- **With quantum physics:** Buhrman, Chandran, Fehr, Gelles, Goyal, Ostrovsky and Schaffner (2014) showed that *any* quantum position check can be beaten with enough entanglement. Their method needs an astronomically large amount.
- **Less is enough:** Beigi and König (2011) cut that down to an exponential amount. That is still huge, but much less. They also showed that some protocols need at least a moderate amount.
- **Bottom line:** perfect, unconditional security is impossible. Any security claim has to assume the cheaters only have a limited amount of entanglement.

D3. How much entanglement breaks each version?

Protocol	What the honest prover handles	Entanglement needed to cheat	Status	Source
Basic BB84 version	1 photon	1 entangled pair	Broken, 100%	Kent, Munro, Spiller 2011; Lau, Lo 2011
BB84 with the basis split into pieces combined by XOR	1 photon	A few pairs (a chain of teleportations)	Broken	Buhrman et al. 2014; the Workbench's "teleport" attack

Protocol	What the honest prover handles	Entanglement needed to cheat	Status	Source
BB84 with a basis from a simple function f	1 photon + 2n ordinary bits	Roughly as much as f's "garden-hose" complexity	Broken with that much	Buhrman, Fehr, Schaffner, Speelman 2013
BB84 with a basis from an f that is easy to compute quantumly	1 photon + 2n ordinary bits	Modest amounts	Broken efficiently	Speelman 2016
BB84 with a basis from a random function f	1 photon + 2n ordinary bits	Proven: cheaters need a number of qubits that grows with n	Secure below that amount	Bluhm, Christandl, Speelman 2022

Open: the proven requirement grows in proportion to n, but the best known attacks need far more (exponential). Nobody knows where the truth lies, or which specific functions f are provably hard.

D4. Noisy or imperfect entanglement

If the shared pair is imperfect (a "Werner state" with quality p between 0 and 1), teleporting adds random errors and the cheaters are right (1 + p)/2 of the time. That only beats the no-entanglement limit of 85% when p is above about 0.71. Below that, cheaters do better without entanglement at all. *(Worked out here for this simple strategy. Cleverer use of noisy entanglement is not fully understood: open.)*

D5. Can you tell an investor "cheaters have limited entanglement"?

Yes, as an engineering assumption with stated limits. Not as a law of physics.

- **Never call the basic BB84 version secure.** One entangled pair, which labs make every day, breaks it.
- **Pitch the version with a function f** and a stated n. The honest device still handles just one photon, while cheaters would need a quantum memory that grows with n, plus split-nanosecond coordination across the site.

- **Say what could weaken it:** large, fast, low-noise quantum memories and networks. State the n and the cheater budget the claim holds against, and update them as technology improves.
- **Wording that stays true:** "secure against any attacker who cannot store and coordinate more than K entangled qubits within the response time", with K taken from the proof.

How to defend against section D

- **Use the version with a hard function f** and a large n. Cost: the prover must compute f within the allowance, and the proofs only say what happens for large n.
- **Shorten the response window.** Cost: faster hardware; the window also sets the precision anyway.
- **Accept a conditional guarantee** and put a number on K.

Trying section D in the Workbench

- **In the Scenarios gallery:** D1 teleportation, the D3 chain in 2D, and D4 with poor and with good entanglement.
- **Build your own:** set "Who answers" to the teleportation attack, then add the twist "Noise on shared entangled pairs". The Workbench's noise setting is 1 − p, so quality 0.9 is noise 0.1.
- **Not simulable in general:** the best attacks on the random-function version need entanglement that grows exponentially, so only very small n fit in a simulator.

E. Special locations, and who is really there

Position checking proves *where* an answer came from, never *who* sent it. Anyone standing at the right spot passes. To also check identity you need a secret key on top, and even that can't stop a prover who willingly shares the key.

#	Situation	Cheater passes per round	Safe?	What it tests	Fix
E1	Cheater standing right where P is	100%	Location yes, identity no	Location vs identity	Add a secret key to the answer
E2	Honest prover helps a cheater	100%	No, and nothing can prevent it	Honest prover	Trust the prover, or lock the key in hardware

#	Situation	Cheater passes per round	Safe?	What it tests	Fix
E3	Someone in the middle delays, replays or tampers	0% (delay or replay); up to the C or D limits if they grab the photon	Yes, but they can block the check	Channel integrity	Repeat rounds, monitor the links
E4	Cheater at or near a verifier	85% near V0, 50% near V1	Yes, unless inside the verifier's building	Trusted verifiers	Secure the verifier sites
E5	Several honest provers close together	Whoever gets the photon passes	Can't tell them apart	Location vs identity	Separate rounds or keys per prover
E6	An answer that seems to come from the right direction	Only within the C or D limits	Yes	Speed of light	Nothing extra: direction is never trusted

E1. A cheater standing exactly where P is

It passes every round. The verifiers only see a correct bit arriving on time, and that is all they check. The QPV Workbench shows this too: a cheater placed at the claimed spot is accepted, with the same safety margin as the honest prover. If both stand there, only one can get the photon, because it can't be copied, and whoever has it answers.

- **Fix: tie the answer to a key.** The real prover holds a secret key shared with the verifiers. Its answer includes a short code that only that key can produce, computed within the same allowance. Buhrman et al. (2014) describe this kind of "position-based authentication".
- **Cost:** keys have to be handed out and stored safely, and the code must be computed fast enough. The guarantee becomes "someone holding the key is at P".

E2. The honest prover helps a cheater

A prover who wants to cheat can give its key to an accomplice standing at P, or let the accomplice answer for it. No protocol can stop this, because the prover's cooperation *is* the attack. Position checking assumes provers want to prove their *own* location.

- **Partial fix:** keep the key inside tamper-proof hardware that only works within a sealed device. Cost: you now have to trust that hardware.

E3. Someone in the middle of the road

- **Delaying messages:** answers arrive late and are rejected. That blocks the check but never fools it.

- **Replaying old answers:** x and θ are new every round, so an old answer is only right half the time.
- **Changing θ on the way:** the honest prover reads with the wrong setting and makes mistakes, which the verifiers notice.
- **Grabbing the photon:** the person in the middle becomes the first cheater of sections C and D, and the same limits apply.

E4. A cheater at or near a verifier

Standing next to V0 is just the "one cheater on V0's side" case (C1): at most 85% per round. Next to V1 it is worth 50%. *Inside* a verifier's building is a different story: a cheater there could read θ or x early and win every time (F4).

E5. Several honest provers close together

They are all inside the same small zone, so timing can't tell them apart. Only one of them can receive and read a given photon. Run separate rounds for each prover with separate keys (E1), or keep claims further apart than $c \times \tau$.

E6. An answer from the right direction, but made elsewhere

Verifiers never use the direction an answer comes from, only its timing and content. Any such trick is one of the team attacks in sections C and D, and it is limited by them. Nothing can beat the speed-of-light limits.

Trying section E in the Workbench

- **In the Scenarios gallery:** E1 impostor, E3 with the basis flipped and with a message held up, E4 near a verifier, and E5 with two parties at the spot.
- **Build your own:** "Add a second party at the claimed spot" (choose who gets the photon), or "Put someone in the middle of a path" (pass messages on after a delay, or flip the basis).
- **Not modelled yet:** keys and authentication codes on the answer, the fix for E1.

F. Lost photons, faster routes and leaks

Lost photons are the easiest real-world way to beat the basic protocol. If "nothing arrived" is an allowed answer, cheaters simply say it whenever they guessed wrong, and the check fails once half the photons are lost. A faster route than the verifiers expect, or a leaked basis, breaks it completely.

F1. Saying "nothing arrived" on purpose

The trick. The first cheater guesses the basis and reads the photon straight away. The two cheaters swap notes as usual. If the guess was right, both give the correct bit. If it was wrong, both say "no photon arrived".

What the verifiers see. About half the rounds report "no photon", and every answered round is correct. An honest prover on a link that loses half or more of its photons looks exactly like that, or worse. So the basic protocol breaks once half the photons are lost. *(Worked out here. This weakness is what Qi and Siopsis, 2015, set out to fix.)*

```
         cheater 1 guesses the basis      cheater 2 gets theta early
   guess right (half the time):  both answer the correct bit
   guess wrong (half the time):  both say "no photon"   -> looks like 50%
loss, no mistakes
```

If every round must be answered instead, at least 71% of photons have to arrive (B6).

F2. Versions that cope with lost photons

Protocol	Copes with	Safe against	Source
Basic BB84 version, "nothing arrived" allowed	Losing less than half	Cheaters without entanglement	Worked out here (F1)
Qi and Siopsis' loss-tolerant protocol	Loss beyond the 50% limit (see the paper for details)	Cheaters without entanglement	Qi, Siopsis, PRA 91, 042337 (2015)
Loss-tolerant checking with ordinary lasers	Loss, including lasers that sometimes send two photons	Cheaters without entanglement	Lim, Xu, Siopsis, Chitambar, Evans, Qi, PRA 94, 032315 (2016)
Single-photon loss-tolerant protocol	Lost photons	Cheaters with limited entanglement	Escolà-Farràs, Speelman, PRL 131, 140802 (2023)

Each paper uses its own exact limits and its own assumptions about the cheaters, so read it before quoting a number. A protocol that copes with heavy loss *and* resists cheaters with lots of entanglement, over practical distances, is still being worked on.

F3. A cheater with a faster route

The verifiers' signals might travel through a long fibre detour, slower than light in air, while a cheater has a straight line of sight. The cheater gains the difference in travel time, on the way there and on the way back:

```
\text{time gained} = \frac{\text{fibre length}}{\text{speed in fibre}} -
\frac{\text{straight distance}}{c} \quad \text{each way}
```

That spare time turns into location blur of about c × (time gained) / 2 (see B4). Example: a 12 m fibre detour around a 3 m gap gives the cheater about 50 ns, or roughly 7.5 m of blur. At that point the check proves almost nothing. *(Worked out here.)*

- **Fix:** set every deadline from the straight-line distance at full light speed, and send the honest signals along that straight line through the air. Cost: you need a clear line of sight.

F4. Faulty or dishonest verifiers, and leaked secrets

- **θ leaks early:** the first cheater reads the photon with the right setting the moment it arrives and passes x along. Both cheaters answer correctly, every round.
- **x leaks:** the check is trivially broken.
- **A dishonest verifier** can accept or reject anything it likes. The whole protocol assumes the verifiers are trustworthy and keep their random bits secret.
- **A verifier with a wrong clock or wrong position** shifts the checked spot, as in B3.
- **Fixes:**
  - make x and θ with a quantum random number generator just before sending, and never store them anywhere shared;
  - protect the link between the verifiers with authentication;
  - physically secure the verifier sites;
  - use several verifiers and require most of them to agree, so one faulty verifier can't decide alone.

  Cost: more verifiers and more trusted equipment.

Trying section F in the Workbench

- **In the Scenarios gallery:** F1 selective "no photon", F3 fibre detour and F4 leaked basis.
- **Build your own:** set "Who answers" to the selective "no photon" attack and add "Let the prover report 'no photon'" (F1). Use "Change a verifier's links" with a route length and "cheaters take a straight path at c" (F3). Use "Leak the verifiers' secret to a party", or the strategy *measure on arrival* with the leaked basis (F4).

- **Not modelled:** the loss-tolerant protocols of F2.